

# Atypical bacteria

## *Chlamydia, Mycoplasma and Rickettsia*

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First stage

SGL: 6

# INTRODUCTION

Atypical bacteria represent a heterogeneous group of microorganisms that differ fundamentally from typical bacteria in :

structure,

metabolism,

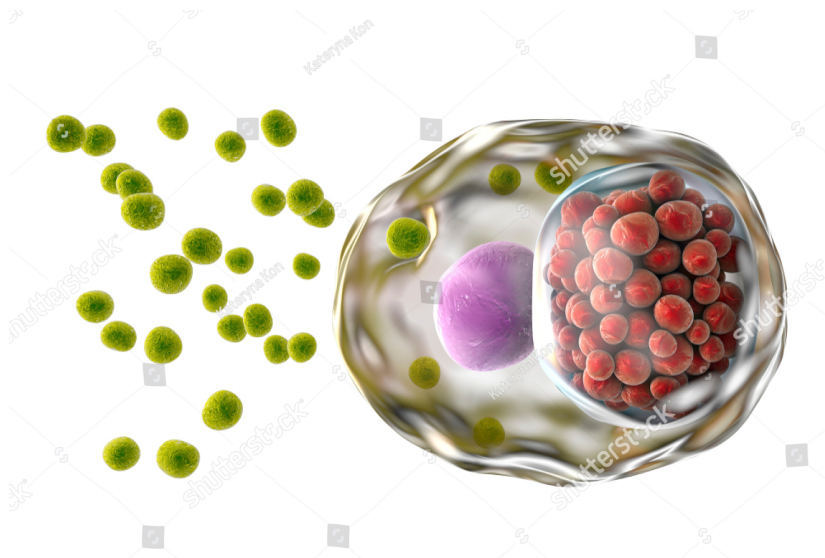
and growth requirements.

These organisms require special diagnostic and therapeutic approaches.

# INTRODUCTION

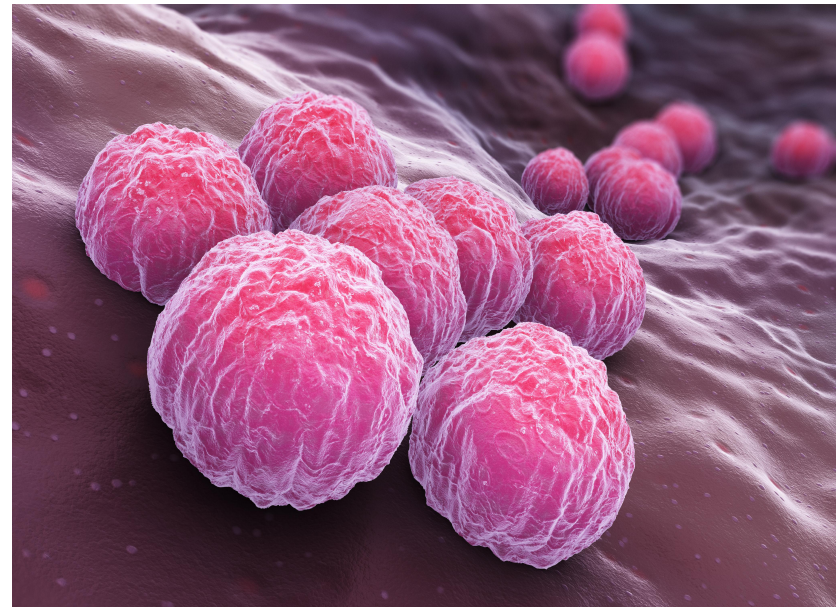
- **Three Major Groups:**
- **Chlamydiae** - Obligate intracellular energy parasites
- **Mycoplasmataceae** - Cell wall-deficient, smallest free-living bacteria
- **Rickettsiaceae** - Obligate intracellular, arthropod-borne pathogens

# Chlamydia



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# *Chlamydia*

- Obligate intracellular gram-negative bacteria
- Cannot grow on artificial media, but can grow on cell line/egg/mice inoculation
- They are filterable and produce inclusion bodies like viruses
- But they differ from viruses being possessing both RNA and DNA.

# *Chlamydia*

- Cell wall **lacks muramic acid** (modified peptidoglycans)
- **Cannot** produce their own **ATP**: They are called Energy parasite as they **depend** on **host cell** ATP.
- Life cycle: They exist in two distinct morphological forms: Elementary body (**EB**) and Reticulate body(**RB**).

# Life cycle of *Chlamydia*

- Chlamydia have two distinct morphological forms:
- The elementary body (EB): an infectious extra-cellular particle (300–400 nm).
- 
- The reticulate body (RB): an intracellular non-infectious particle (800–1,000 nm).

# Life cycle of *Chlamydia*

- EBs bind to specific host cell receptors and enter by endocytosis.
- Target cells include
  - conjunctival
  - urethral
  - rectal
  - endocervical epithelial cells.



# Life cycle of *Chlamydia*

- They multiply in the cytoplasm of host cells after a well-defined developmental cycle:
- Intracellular EBs remain within phagosomes and replicate.
- Lysosomal fusion with the phagosome containing EBs is inhibited, probably by chlamydial cell-wall components.

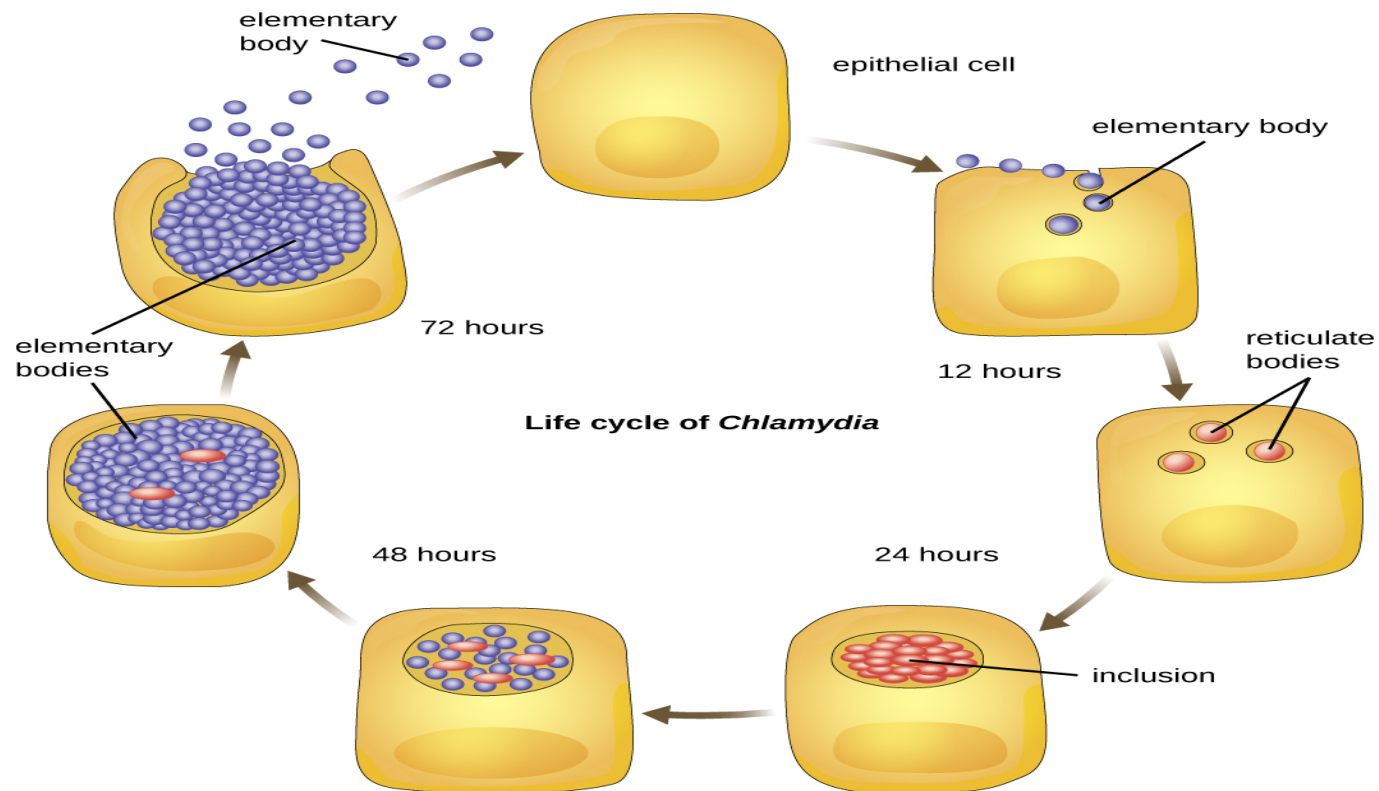
# Life cycle of *Chlamydia*

Inside host cell, EBs become **metabolically active** and form **RBs** that **synthesise DNA, RNA and protein**, utilising **host cell** adenosine triphosphate (**ATP**) as an energy source.

**RBs** undergo multiple divisions and the phagosome becomes an '**inclusion body**' (visible by light microscopy).

The **RBs** reorganize into smaller **EBs** and, after a further, the host cell lyses and infective EBs are **released**.

# Life cycle of *Chlamydia*



# Genera of Medical Importance

- Genera of Medical Importance
- *Chlamydia trachomatis*
- *Chlamydia pneumoniae* (Chlamydophila pneumoniae)
- *Chlamydia psittaci* (Chlamydophila psittaci)

# Diseases

- ❖ *Chlamydia trachomatis* causes:
  1. **Eye** (conjunctivitis, trachoma),
  2. **Respiratory** (pneumonia),
  3. **Genital tract** (urethritis, lymphogranuloma venereum) infections.
- ❖ *Chlamydia pneumoniae* causes **atypical pneumonia**.
- ❖ *Chlamydia psittaci* causes **psittacosis**

# Laboratory diagnosis

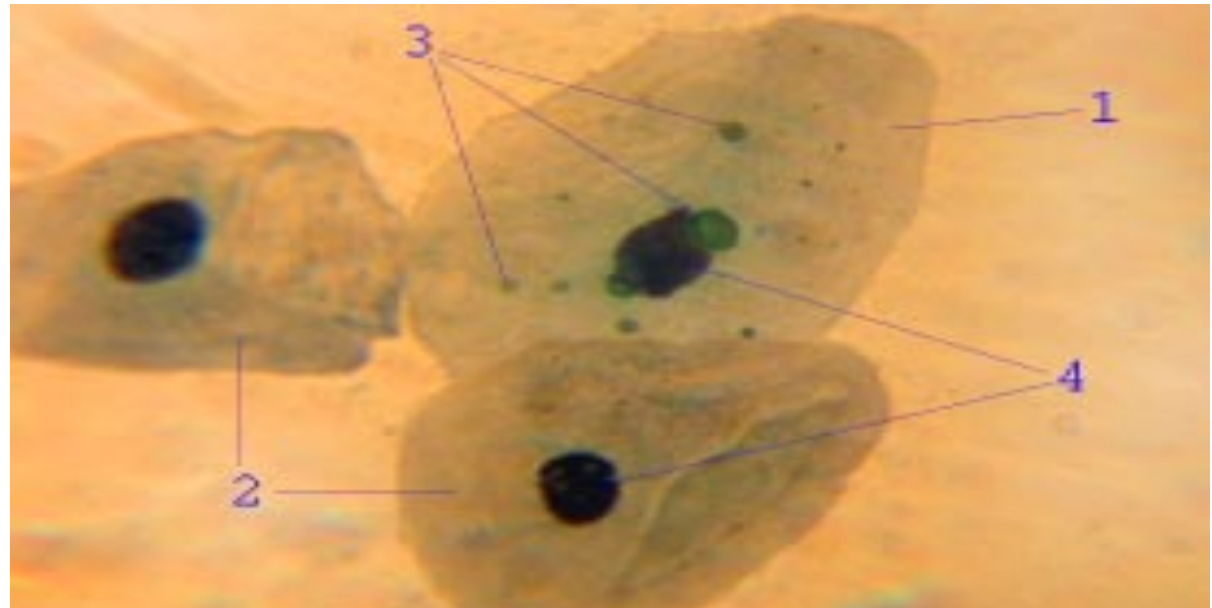
- *Chlamydia* do **not grow** on bacterial **culture media**.
- Cell culture: **swabs from the affected site**, collected in a special transport medium, are inoculated **into tissue culture**.

# Laboratory diagnosis

- Growth of *C. trachomatis* is recognised by **Giemsa staining**, showing intracellular bodies
- Direct antigen detection can be made by immunofluorescence or enzyme-linked immunosorbent assay (**ELISA**) methods.
- Nucleic acid amplification tests to detect either chlamydial DNA or RNA, using polymerase chain reaction (**PCR**)

# *Chlamydia pneumoniae*

- Micrograph of *Chlamydia pneumoniae* in an epithelial cell in acute bronchitis:
- 1 – infected epitheliocyte,
- 2 – uninfected epitheliocytes,
- 3 – chlamydial inclusion bodies in cell,
- 4 – cell nuclei





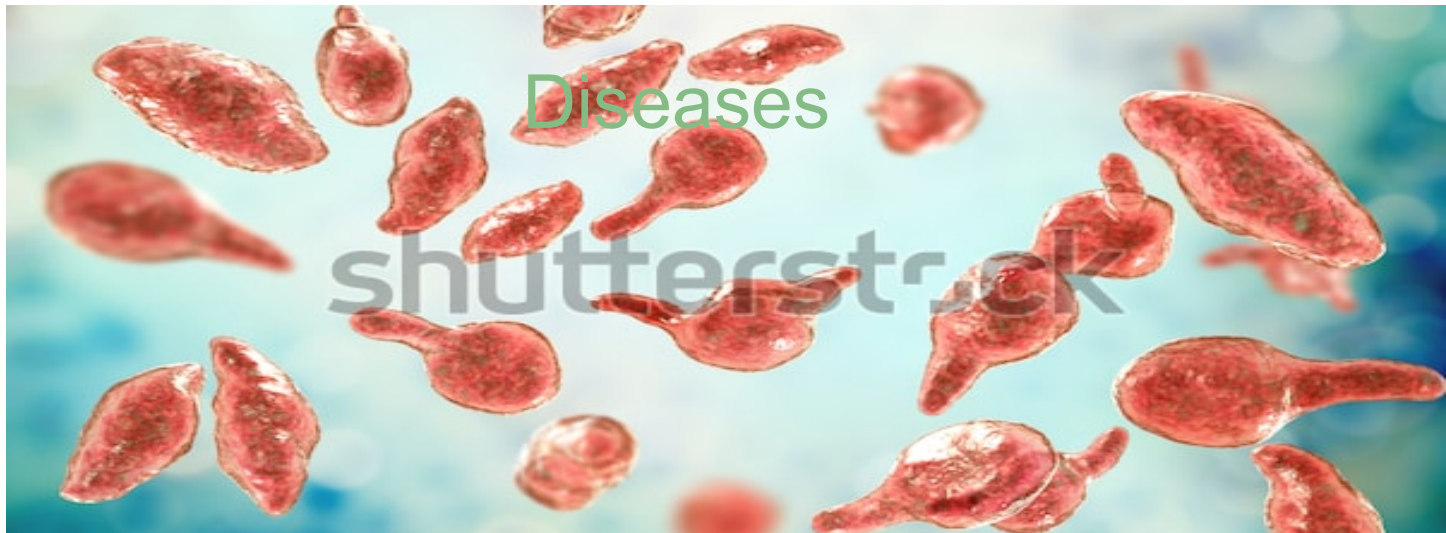
# Chlamydiaceae of Medical Importance

| Species               | Disease                                                                   | Natural Hosts | Mode of Transmission to Humans         | Number of Immunologic Types | Diagnosis                                                                                         | Treatment                 |
|-----------------------|---------------------------------------------------------------------------|---------------|----------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------|---------------------------|
| <i>C. trachomatis</i> | Urethritis, pneumonia, conjunctivitis, lymphogranuloma venereum, trachoma | Humans        | Sexual contact; perinatal transmission | More than 15                | Inclusions in epithelial cells seen with Giemsa stain or by immunofluorescence; also cell culture | Doxycycline, erythromycin |
| <i>C. pneumoniae</i>  | Atypical pneumonia                                                        | Humans        | Respiratory droplets                   | 1                           | Serologic test                                                                                    | Doxycycline               |
| <i>C. psittaci</i>    | Psittacosis (pneumonia)                                                   | Birds         | Inhalation of dried bird feces         | 1                           | Serologic test (cell culture rarely done)                                                         | Doxycycline               |

# Treatment and Prevention

- Treatment—doxycycline or azithromycin
- Prevention
  - ● Erythromycin is effective in infected mothers to prevent neonatal disease.
  - ● Treat neonatal conjunctivitis with erythromycin to prevent pneumonia.

# *Mycoplasma*



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# *Mycoplasma* Genus Features

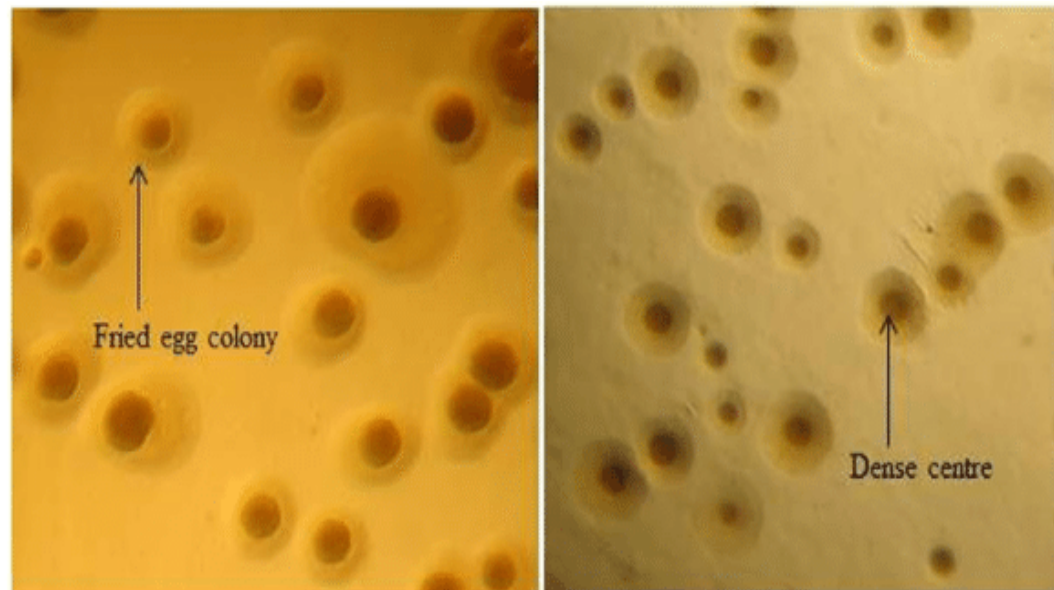
- **Smallest free-living** (extracellular) bacteria
  - Small (0.15–0.25 $\mu\text{m}$ )
  - Gram-negative microorganisms;
  - Pleomorphic with no cell wall (**missing peptidoglycan**)
  - Grow slowly on enriched media (**require cholesterol for in vitro culture**)
  - Aerobic or facultatively anaerobic
- Sterols in membrane**

# Diseases

- **Respiratory:** pharyngitis; community-acquired pneumonia (*M. pneumoniae*)
- **Central nervous system:** meningitis, encephalitis, transverse myelitis (*M. pneumoniae*)
- **Genitourinary:** urethritis, pelvic inflammatory disease (*M. hominis*, *M. genitalium*: urethritis)

# Laboratory diagnosis

- Culture: *mycoplasmas* can be grown on enriched media; penicillin is often added to inhibit other microorganisms.
- Producing colonies with a 'fried egg' appearance.



# Laboratory diagnosis

## Serology

- Serological tests:
- Complement fixation test
- Enzyme immunoassays

# Treatment and Prevention

- Treatment—erythromycin, azithromycin, clarithromycin;
- no cephalosporins or penicillins
- Prevention—none



# *Rickettsia* (small red) inside human cell, Gram-negative bacteria



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# *Rickettsia* Genus Features

- Aerobic, gram-negative bacilli (0.2–1  $\mu\text{m}$  diameter)
- Obligate intra cellular pathogens; utilise ATP from host cell
- Grow only in tissue culture

# Rickettsia



## Species of Medical Importance

- *Rickettsia rickettsii*
- *Rickettsia prowazekii*
- *Rickettsia typhi*
- *Orientia tsutsugamushi* (formerly *R. tsutsugamushi*)
- *Ehrlichia spp.*

# *Rickettsia*

Rickettsial infections are **zoonoses** with a variety of animal reservoirs and insect vectors.

Rickettsiaceae cause a number of important human infections, including **typhus and the related spotted fevers**.

# Infections Caused by Rickettsiae and Close Relatives

| Group Disease                | Bacterium                                        | Arthropod Vector | Reservoir Host       |
|------------------------------|--------------------------------------------------|------------------|----------------------|
| Rocky Mountain Spotted Fever | <i>R. rickettsii</i>                             | Ticks            | Ticks, dogs, rodents |
| Epidemic Typhus              | <i>R. prowazekii</i>                             | Human louse      | Humans               |
| Endemic Typhus               | <i>R. typhi</i>                                  | Fleas            | Rodents              |
| Scrub Typhus                 | <i>Orientia tsutsugamushi</i>                    | Mites            | Rodents              |
| Ehrlichiosis                 | <i>E. chafeensis</i><br><i>E. phagocytophila</i> | Tick             | Small mammals        |

# Laboratory diagnosis

- This is by serology.
- Serological immunofluorescence assays (IFA)
- ELISA
- Weil-Felix test

# Treatment and Prevention

- Treatment—doxycycline
- Prevention—tick protection and prompt removal of tick
- doxycycline effective in exposed persons

## Comparison of the Chlamydiaceae, Rickettsiaceae, and Mycoplasmataceae with Typical Bacteria

|                                       | Typical Bacteria ( <i>S. aureus</i> ) | Chlamydiaceae           | Rickettsiaceae       | Mycoplasmataceae |
|---------------------------------------|---------------------------------------|-------------------------|----------------------|------------------|
| Obligate intracellular parasite?      | Mostly no                             | Yes                     | Yes                  | No               |
| Make ATP?                             | Normal ATP                            | No ATP                  | Limited ATP          | Normal ATP       |
| Peptidoglycan layer in cell envelope? | Normal peptidoglycan                  | Modified* peptidoglycan | Normal peptidoglycan | No peptidoglycan |

\*Chlamydial peptidoglycan lacks muramic acid and is considered by some as modified, by others as absent.



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